intervention. For example, AI-enabled swarm drones have been successfully used for perimeter defense, target tracking, and rapid response missions in complex terrains. Their capability to share sensor data in real-time improves situational awareness and reduces reaction time in critical missions.

Impact: Enhanced mission precision, reduced human casualties, and efficient utilization of defense resources.

6.7.2 Disaster response and humanitarian aid

During natural disasters such as earthquakes, floods, or wildfires, fifth-generation drones equipped with thermal imaging and LiDAR sensors have played a crucial role in locating survivors, assessing structural damage, and delivering emergency supplies. Autonomous navigation systems enable these drones to operate in GPS-denied environments, ensuring effective coordination even under extreme conditions.

Impact: Rapid disaster assessment, minimized rescue delays, and improved resource deployment efficiency.

6.7.3 Precision agriculture and environmental monitoring

The application of fifth-generation drones in agriculture has transformed crop monitoring, irrigation management, and pest control. Equipped with hyperspectral cameras and AI-driven analytics, these drones can detect nutrient deficiencies, soil moisture levels, and disease outbreaks. In environmental applications, they assist in biodiversity mapping, pollution tracking, and forest fire prediction through real-time data collection and predictive algorithms. *Impact*: Increased agricultural productivity, reduced input costs, and sustainable resource management.